

# FACTSHEET FS12 · Buy at the 52-week low

**What changed on 2 October 2026.** Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS\_RERUN\_2026-10-02.md.

**US, 1999–2012 (pre-registered second period, run 2 October 2026).** The same rules on Sharadar's point-in-time S&P 500 with delisted stocks, 1999–2012, a period this factsheet was not built on. L/S net: t -0.27 (2013–2026: -3.07), same sign, weaker. Long-only alpha vs the equal-weight universe: +1.1% a year, t +0.43 (2013–2026: -2.94), reversed. The 2013–2026 US losses did not repeat; they read as specific to that period. Registered verdicts are unchanged. Pre-registration, method and every result: PREREG\_US\_1999\_2012.md.

## Should you buy stocks at their yearly low?

Henrik Gjerming · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

### STRATEGY

Contrarian / bottom-fishing

### ORIGIN

Market folklore (bottom-fishing); tested against George & Hwang (2004)

### FAMILY

Reversal & bottom-fishing

### UNIVERSES

US · EU · UK · DK · SCANDI · World (point-in-time)

### BACKTEST

Feb 2013 – Aug 2026, monthly, net of costs

### BOOKS

Long/short and long-only (stocks closest to their 52-week low)

### REBALANCE

Monthly

### STATUS

Descriptive factsheet, not a registered trial

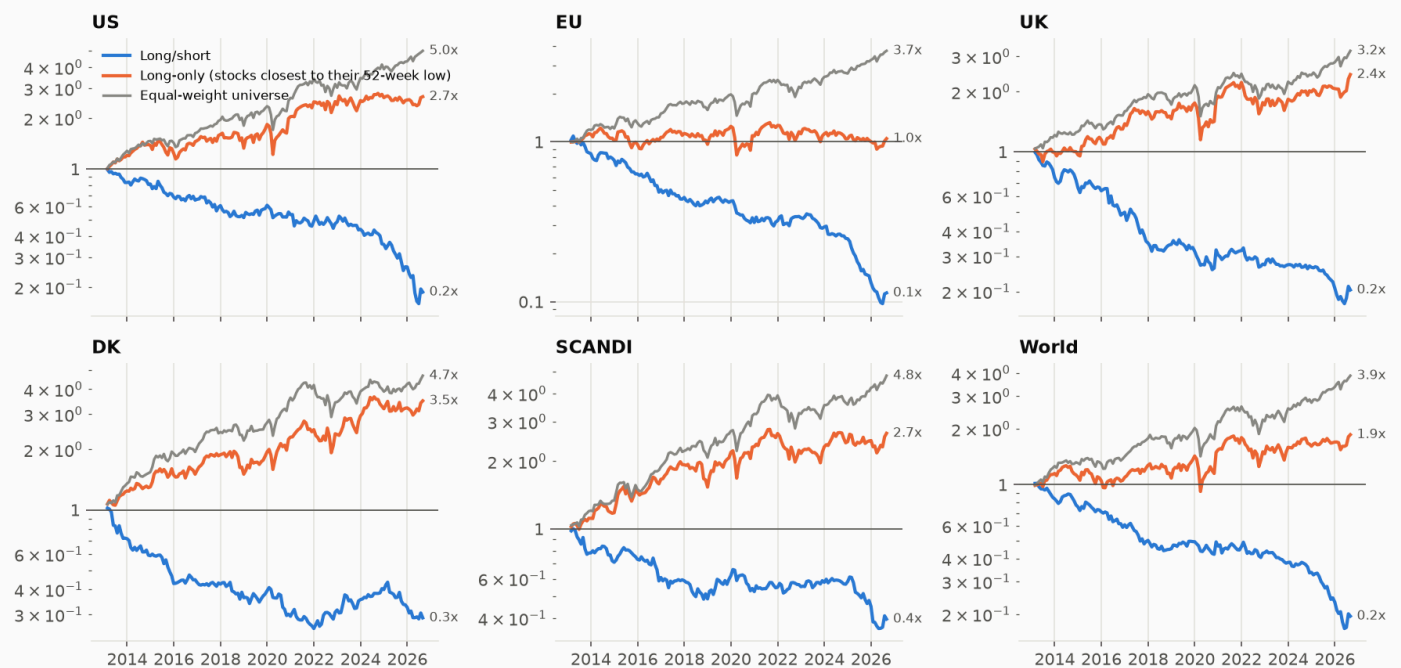
**The claim.** Folklore says stocks at their yearly low are bargains: 'buy low'. The academic evidence points the other way: stocks near their lows tend to keep underperforming (George & Hwang 2004), and in the series A10 Free Fallin' found knives rebound only in market-wide crashes. **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned -14.7% to -5.8% a year and was positive in none of the six; no universe passes the 2.87 gate (significantly negative in US, EU, World). The long-only book (stocks closest to their 52-week low) had a higher Sharpe than the equal-weight universe in none of the six, with alphas of -9.7% to -2.5%, none past the gate. **What goes wrong (§10):** no single dominant drag. None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Reversal & bottom-fishing; its returns sit closest to the momentum cluster of the signal library. **360° view (§12):** nearest neighbours Momentum 12-1, Residual momentum; after those neighbours and the market, nothing is left (alpha  $|t| < 2$  everywhere): the signal is redundant.

# 1. Headline performance

| Universe | Names / groups | L/S return / yr | L/S t | L/S CAPM alpha (t) | L/S beta | Long-only CAGR | Long-only Sharpe | Long-only max DD | EW CAGR | EW Sharpe | Size-weighted CAGR |
|----------|----------------|-----------------|-------|--------------------|----------|----------------|------------------|------------------|---------|-----------|--------------------|
| US       | 502 / 10       | -10.8%          | -3.07 | -12.4% (-3.9)      | 0.13     | +7.5%          | 0.47             | -34%             | +12.6%  | 0.85      | +14.9%             |
| EU       | 264 / 10       | -14.7%          | -3.47 | -15.8% (-3.7)      | 0.10     | +0.3%          | 0.11             | -34%             | +10.1%  | 0.75      | +10.2%             |
| UK       | 327 / 10       | -10.4%          | -2.64 | -13.2% (-3.5)      | 0.30     | +6.8%          | 0.44             | -35%             | +8.9%   | 0.68      | +7.8%              |
| DK       | 20 / 3         | -7.9%           | -1.96 | -9.0% (-2.2)       | 0.09     | +9.7%          | 0.60             | -31%             | +12.0%  | 0.83      | +10.2%             |
| SCANDI   | 70 / 5         | -5.8%           | -1.62 | -7.3% (-2.0)       | 0.12     | +7.5%          | 0.51             | -29%             | +12.2%  | 0.89      | +11.0%             |
| World    | 1089 / 10      | -11.0%          | -3.27 | -13.0% (-4.2)      | 0.18     | +4.7%          | 0.33             | -36%             | +10.5%  | 0.71      | +10.3%             |

Monthly, local currency (World in USD). L/S = equal-weighted stocks closest to their 52-week low minus stocks furthest above it, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

## Buy at the 52-week low on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjerner · Project 10 factsheets

## 2. Strategy description

Folklore says stocks at their yearly low are bargains: 'buy low'. The academic evidence points the other way: stocks near their lows tend to keep underperforming (George & Hwang 2004), and in the series A10 Free Fallin' found knives rebound only in market-wide crashes.

**Why it might work.** Folklore: mean reversion and value. Evidence: momentum and anchoring (George & Hwang 2004) predict continued weakness; losers carry high beta and volatility.

| Rule      | Original (Market folklore (bottom-fishing); tested against George & Hwang (2004)) | This factsheet                                               |
|-----------|-----------------------------------------------------------------------------------|--------------------------------------------------------------|
| Signal    | Folklore: buy at or near the 52-week low                                          | Total-return price / 252-day low; lowest = at the low        |
| Portfolio | No standard rule                                                                  | Extreme quantile L/S (at the low minus far above), one month |
| Weighting | —                                                                                 | Equal weighted                                               |

## 3. Data load

| Universe | Prices                                                                  | Membership                                                                                                                                                                                                                | Names eligible (median) | Currency                   |
|----------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------|
| US       | Sharadar SEP ccloseadj (total return), 805 tickers incl. delisted       | <b>Point-in-time:</b> the S&P 500 members at each month-end (Sharadar add/remove history; corrected 2 Oct 2026)                                                                                                           | ~503                    | USD                        |
| EU       | Project1 yfinance cache, Adj Close (TR)                                 | <b>Point-in-time:</b> union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced | ~238                    | local (EUR, SEK, DKK, PLN) |
| UK       | Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×) | <b>Point-in-time:</b> FTSE 100 + FTSE 250, quarterly snapshots; 75% priced                                                                                                                                                | ~242                    | GBP                        |
| DK       | Project1 cache, Adj Close                                               | <b>Point-in-time:</b> OMX Copenhagen 25, quarterly snapshots; 91% priced                                                                                                                                                  | ~19                     | DKK                        |
| SCANDI   | Project1 cache, Adj Close                                               | <b>Point-in-time:</b> OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced                                                                                                                                 | ~62                     | local (DKK, SEK, EUR)      |
| World    | US Sharadar + UK + EU above                                             | <b>Point-in-time</b> union of the three; local-currency returns summed, not converted                                                                                                                                     | ~963                    | mixed                      |

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as run\_wiki\_pit.py / run\_eu\_pit.py), with a short list of verified ticker renames (e.g. NZYM-B → NSIS-B, WDH → DEMANT, DAI → MBG). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-

quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

## 4. Signal creation

$LO52_{i,t} = P_{i,t} / \min(P_{i,t-251..t})$ ; the long leg holds the stocks with the lowest value (at the low). Signals use only information up to month-end  $t$  and are traded in month  $t+1$ . Eligibility: in the point-in-time universe on any of the last five trading days of month  $t$ .

## 5. Model build

- Sort eligible stocks into equal-count groups (deciles  $\geq 100$  names, quintiles 50–99, terciles  $< 50$ ; fixed per universe).
- **Long/short** = stocks closest to their 52-week low minus stocks furthest above it; **long-only** = stocks closest to their 52-week low; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

## 6. Performance in detail

US (S&P 500 members, point-in-time), median 502 names, 10 groups

| Book                                        | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]      | Sortino | Max DD | Hit | t(mean) | Beta | Alpha/yr (t)  | CVaR 95% (month) |
|---------------------------------------------|-----------|--------|-------|----------------------|---------|--------|-----|---------|------|---------------|------------------|
| L/S (net)                                   | -10.8%    | -11.6% | 17.3% | -0.62 [-1.03, -0.21] | -0.84   | -84%   | 42% | -3.07   | 0.13 | -12.4% (-3.9) | 11.8%            |
| Long-only (net)                             | +9.2%     | +7.5%  | 19.4% | 0.47 [0.10, 0.92]    | 0.61    | -34%   | 60% | 2.54    | 1.14 | -5.7% (-2.9)  | 11.3%            |
| Stocks closest to their 52-week low (gross) | +10.3%    | +8.7%  | 19.5% | 0.53 [0.16, 0.98]    | 0.71    | -33%   | 60% | 2.87    | 1.14 | -4.6% (-2.4)  | 11.1%            |
| Stocks furthest above it (gross)            | +18.6%    | +18.2% | 19.2% | 0.97 [0.56, 1.41]    | 1.72    | -23%   | 61% | 4.85    | 1.01 | +5.4% (2.6)   | 10.1%            |
| EW universe                                 | +13.1%    | +12.6% | 15.5% | 0.85 [0.41, 1.42]    | 1.29    | -28%   | 67% | 4.20    | 1.00 | –             | 9.2%             |

| Book                   | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]   | Sortino | Max DD | Hit | t(mean) | Beta | Alpha/yr (t) | CVaR 95% (month) |
|------------------------|-----------|--------|-------|-------------------|---------|--------|-----|---------|------|--------------|------------------|
| Size-weighted universe | +14.9%    | +14.9% | 14.2% | 1.05 [0.60, 1.61] | 1.73    | -24%   | 69% | 5.29    | 0.86 | +3.7% (2.4)  | 8.3%             |

EU (11 national blue-chip indices, point-in-time), median 264 names, 10 groups

| Book                                        | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]      | Sortino | Max DD | Hit | t(mean) | Beta | Alpha/yr (t)  | CVaR 95% (month) |
|---------------------------------------------|-----------|--------|-------|----------------------|---------|--------|-----|---------|------|---------------|------------------|
| L/S (net)                                   | -14.7%    | -14.8% | 15.3% | -0.96 [-1.59, -0.42] | -1.13   | -91%   | 38% | -3.47   | 0.10 | -15.8% (-3.7) | 9.4%             |
| Long-only (net)                             | +1.8%     | +0.3%  | 17.3% | 0.11 [-0.32, 0.56]   | 0.03    | -34%   | 53% | 0.47    | 1.08 | -9.7% (-5.0)  | 10.3%            |
| Stocks closest to their 52-week low (gross) | +3.0%     | +1.5%  | 17.2% | 0.17 [-0.26, 0.63]   | 0.13    | -34%   | 53% | 0.76    | 1.07 | -8.6% (-4.4)  | 10.2%            |
| Stocks furthest above it (gross)            | +15.2%    | +14.7% | 17.1% | 0.89 [0.37, 1.47]    | 1.48    | -32%   | 63% | 3.28    | 0.98 | +4.7% (1.6)   | 8.6%             |
| EW universe                                 | +10.7%    | +10.1% | 14.3% | 0.75 [0.29, 1.30]    | 1.14    | -26%   | 62% | 3.18    | 1.00 | –             | 8.4%             |
| Size-weighted universe                      | +10.6%    | +10.2% | 13.7% | 0.77 [0.32, 1.29]    | 1.21    | -25%   | 58% | 3.37    | 0.92 | +0.8% (0.6)   | 7.9%             |

UK (FTSE 350, point-in-time), median 327 names, 10 groups

| Book                                        | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]      | Sortino | Max DD | Hit | t(mean) | Beta | Alpha/yr (t)  | CVaR 95% (month) |
|---------------------------------------------|-----------|--------|-------|----------------------|---------|--------|-----|---------|------|---------------|------------------|
| L/S (net)                                   | -10.4%    | -11.1% | 16.2% | -0.64 [-1.16, -0.16] | -0.91   | -83%   | 38% | -2.64   | 0.30 | -13.2% (-3.5) | 9.6%             |
| Long-only (net)                             | +8.3%     | +6.8%  | 18.9% | 0.44 [0.01, 0.92]    | 0.59    | -35%   | 53% | 1.87    | 1.18 | -3.0% (-1.4)  | 10.7%            |
| Stocks closest to their 52-week low (gross) | +9.5%     | +8.1%  | 18.9% | 0.50 [0.07, 0.98]    | 0.71    | -34%   | 53% | 2.15    | 1.18 | -1.8% (-0.8)  | 10.6%            |
| Stocks furthest above it (gross)            | +17.5%    | +17.5% | 16.1% | 1.09 [0.57, 1.69]    | 1.86    | -23%   | 64% | 4.21    | 0.88 | +9.0% (3.3)   | 9.0%             |
| EW universe                                 | +9.6%     | +8.9%  | 14.2% | 0.68 [0.20, 1.28]    | 0.96    | -30%   | 58% | 2.87    | 1.00 | –             | 8.8%             |
| Size-weighted universe                      | +8.4%     | +7.8%  | 12.9% | 0.65 [0.16, 1.25]    | 0.90    | -28%   | 62% | 2.68    | 0.84 | +0.3% (0.2)   | 7.8%             |

DK (OMXC25, point-in-time), median 20 names, 3 groups



| Book                                        | Mean / yr | CAGR   | Vol   | Sharpe [95% CI]    | Sortino | Max DD | Hit | t(mean) | Beta | Alpha/yr (t) | CVaR 95% (month) |
|---------------------------------------------|-----------|--------|-------|--------------------|---------|--------|-----|---------|------|--------------|------------------|
| Stocks closest to their 52-week low (gross) | +7.7%     | +5.9%  | 19.7% | 0.39 [-0.03, 0.84] | 0.47    | -35%   | 58% | 1.78    | 1.15 | -5.3% (-3.4) | 11.0%            |
| Stocks furthest above it (gross)            | +16.1%    | +15.6% | 17.6% | 0.92 [0.43, 1.49]  | 1.51    | -27%   | 58% | 3.76    | 0.97 | +5.2% (2.7)  | 9.9%             |
| EW universe                                 | +11.3%    | +10.5% | 16.0% | 0.71 [0.25, 1.28]  | 1.02    | -30%   | 63% | 3.07    | 1.00 | –            | 9.7%             |
| Size-weighted universe                      | +11.0%    | +10.3% | 14.8% | 0.74 [0.28, 1.31]  | 1.10    | -28%   | 61% | 3.19    | 0.91 | +0.7% (0.8)  | 8.9%             |

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2013 | -16.8% | -15.6% | -25.7% | -28.2% | -21.9% | -16.2% |
| 2014 | -2.1%  | -9.1%  | -11.1% | -17.8% | -9.1%  | -9.5%  |
| 2015 | -16.4% | -17.9% | -3.5%  | -27.4% | +3.0%  | -5.9%  |
| 2016 | -2.8%  | -20.3% | -21.9% | +1.3%  | -17.0% | -14.8% |
| 2017 | -8.2%  | -10.5% | -29.8% | -2.0%  | -3.2%  | -17.6% |
| 2018 | -15.1% | -7.6%  | -1.6%  | -14.4% | -17.0% | -9.8%  |
| 2019 | +18.3% | +2.5%  | -7.6%  | +8.2%  | +26.8% | +9.0%  |
| 2020 | -12.0% | -26.5% | -4.1%  | -25.3% | -15.0% | -5.3%  |
| 2021 | -8.4%  | +2.5%  | +5.1%  | -13.1% | +4.5%  | -0.7%  |
| 2022 | -1.1%  | +4.4%  | -15.1% | +23.7% | +0.6%  | -7.1%  |
| 2023 | -9.6%  | -12.3% | +0.1%  | +12.7% | +4.6%  | -11.5% |
| 2024 | -18.7% | -15.7% | -3.0%  | +12.2% | +0.8%  | -11.6% |
| 2025 | -26.4% | -46.9% | -21.4% | -18.5% | -22.5% | -30.9% |
| 2026 | -29.0% | -13.6% | -2.1%  | -11.4% | -12.1% | -16.0% |

Calendar-year returns, long-only (net)

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2013 | +20.5% | +11.7% | -0.7%  | +24.3% | +11.4% | +14.9% |
| 2014 | +14.6% | -8.7%  | -2.8%  | +4.8%  | +5.8%  | -4.3%  |
| 2015 | -10.1% | -5.4%  | +19.1% | +14.4% | +23.4% | -4.5%  |
| 2016 | +18.6% | +10.8% | +20.2% | +5.6%  | +14.2% | +4.2%  |
| 2017 | +8.8%  | +5.7%  | +15.2% | +19.7% | +16.0% | +16.2% |
| 2018 | -17.1% | -14.5% | -8.2%  | -20.1% | -20.4% | -14.7% |

| Year | US     | EU     | UK     | DK     | SCANDI | World  |
|------|--------|--------|--------|--------|--------|--------|
| 2019 | +38.2% | +29.6% | +20.2% | +29.3% | +42.9% | +31.6% |
| 2020 | +11.2% | -13.2% | +8.2%  | +18.5% | +6.0%  | +8.0%  |
| 2021 | +22.2% | +16.8% | +17.1% | +8.9%  | +14.4% | +16.3% |
| 2022 | -3.6%  | -11.4% | -22.9% | -5.4%  | -18.5% | -18.2% |
| 2023 | +5.7%  | -0.1%  | +6.8%  | +23.9% | +7.2%  | +8.3%  |
| 2024 | +4.0%  | -6.7%  | +8.8%  | +8.4%  | +5.2%  | +1.3%  |
| 2025 | -3.1%  | -4.1%  | +3.8%  | +1.0%  | +1.2%  | +5.3%  |
| 2026 | +4.6%  | +4.4%  | +17.6% | +8.5%  | +7.7%  | +9.6%  |

## 7. Trading record

The rebalance log per universe is in `results/xs_FS12_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

| Universe | Rebalances | Names long / short | Turnover long leg / month | Turnover short leg / month | L/S months positive | Worst L/S month | When    |
|----------|------------|--------------------|---------------------------|----------------------------|---------------------|-----------------|---------|
| US       | 163        | 51 / 50            | 47%                       | 25%                        | 42%                 | -21.1%          | 2026-04 |
| EU       | 163        | 27 / 26            | 47%                       | 25%                        | 38%                 | -12.0%          | 2022-03 |
| UK       | 163        | 33 / 32            | 50%                       | 23%                        | 38%                 | -15.3%          | 2016-04 |
| DK       | 163        | 7 / 6              | 22%                       | 20%                        | 46%                 | -14.1%          | 2013-05 |
| SCANDI   | 163        | 14 / 14            | 31%                       | 24%                        | 42%                 | -14.7%          | 2026-01 |
| World    | 163        | 109 / 108          | 49%                       | 25%                        | 39%                 | -11.9%          | 2026-04 |

**Current book** (signal at 31 August 2026; first 8 names per leg)

| Universe | Long: stocks closest to their 52-week low                                  | Short: stocks furthest above it                                                    |
|----------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| US       | VICI, APTV, LII, MLM, LVS, PCG, PNR, NRG                                   | SNDK, MU, LITE, MRNA, WDC, STX, DELL, CIEN                                         |
| EU       | AD.AS, BPOST.BR, EL.PA, VNA.DE, UMG.AS, BAR.BR, ELISA.HE, KER.WA           | KGH.WA, ASML.AS, MT.AS, MTS.MC, TYRES.HE, STMPA.PA, NOKIA.HE, STMMI.MI             |
| UK       | AML.L, SN.L, BAG.L, ENT.L, RTO.L, IMB.L, PRN.L, SAFE.L                     | CWR.L, CMCX.L, SAGA.L, SSIT.L, HAS.L, CCC.L, KLR.L, TRST.L                         |
| DK       | TRYG.CO, DSV.CO, RBREW.CO, GN.CO, AMBU-B.CO, COLO-B.CO, CARL-B.CO, BAVA.CO | PNDORA.CO, VWS.CO, MAERSK-B.CO, MAERSK-A.CO, JYSK.CO, DEMANT.CO, ISS.CO, DANSKE.CO |

| Universe | Long: stocks closest to their 52-week low                                  | Short: stocks furthest above it                                                   |
|----------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| SCANDI   | ELISA.HE, AZN.ST, TRYG.CO, KEMIRA.HE, GN.CO, RBREW.CO, KNEBV.HE, KALMAR.HE | TYRES.HE, NOKIA.HE, NESTE.HE, MAERSK-B.CO, QTCOM.HE, MAERSK-A.CO, VWS.CO, ZEAL.CO |

## 8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

### Long/short

| Universe | Factor model           | Alpha / yr | t(alpha) | R <sup>2</sup> | Loadings with  t  ≥ 2 (t)                                                                        |
|----------|------------------------|------------|----------|----------------|--------------------------------------------------------------------------------------------------|
| US       | JKP USA 7 themes       | -5.6%      | -2.32    | 0.64           | momentum -1.35 (-11.2), low_risk +0.65 (+5.9), short_term_reversal +0.68 (+2.6)                  |
| EU       | French Europe 5F + WML | -5.2%      | -1.39    | 0.33           | RMW +0.70 (+2.2), WML -0.95 (-6.6)                                                               |
| UK       | JKP GBR 7 themes       | -7.7%      | -2.22    | 0.54           | mkt +0.18 (+3.5), momentum -1.11 (-7.0), low_risk +0.67 (+3.5), short_term_reversal +0.78 (+5.7) |
| DK       | JKP DNK 7 themes       | +1.2%      | 0.34     | 0.48           | momentum -1.02 (-8.7), low_risk +0.63 (+5.8)                                                     |
| SCANDI   | French Europe 5F + WML | +0.7%      | 0.17     | 0.15           | WML -0.60 (-3.6)                                                                                 |
| World    | JKP World 7 themes     | -9.7%      | -2.70    | 0.34           | mkt +0.17 (+3.2), momentum -0.83 (-6.7), low_risk +0.43 (+2.3)                                   |

### Long-only

| Universe | Factor model           | Alpha / yr | t(alpha) | R <sup>2</sup> | Loadings with  t  ≥ 2 (t)                                                                                         |
|----------|------------------------|------------|----------|----------------|-------------------------------------------------------------------------------------------------------------------|
| US       | JKP USA 7 themes       | +1.0%      | 0.61     | 0.92           | mkt +0.95 (+19.2), size -0.24 (-3.1), value +0.33 (+3.8), momentum -0.57 (-7.8), short_term_reversal +0.54 (+3.6) |
| EU       | French Europe 5F + WML | +0.8%      | 0.32     | 0.76           | Mkt-RF +0.74 (+9.3), WML -0.57 (-8.0)                                                                             |
| UK       | JKP GBR 7 themes       | +4.2%      | 1.53     | 0.77           | mkt +0.64 (+14.8), momentum -0.49 (-3.2), low_risk -0.41 (-2.1), short_term_reversal +0.77 (+4.7)                 |
| DK       | JKP DNK 7 themes       | +7.2%      | 2.45     | 0.68           | mkt +0.74 (+11.1), momentum -0.56 (-5.7)                                                                          |
| SCANDI   | French Europe 5F + WML | +7.8%      | 2.46     | 0.59           | Mkt-RF +0.68 (+7.6), RMW +0.48 (+2.1), WML -0.44 (-3.7)                                                           |
| World    | JKP World 7 themes     | +0.0%      | 0.01     | 0.82           | mkt +0.81 (+15.8), value +0.29 (+2.5), momentum -0.39 (-4.7)                                                      |

## 9. Statistical verdict

- **Gate:**  $|t| > 2.87$  (2 books  $\times$  6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in none of the six; passes in 0 of 6.
- **L/S CAPM alpha:** -15.8% to -7.3%,  $|t|$  past the gate in 4 of 6.
- **Long-only:** Sharpe above the equal-weight universe in none of the six; alpha -9.7% to -2.5%, passing in 0 of 6.
- **Probabilistic Sharpe** of the L/S,  $P(\text{true Sharpe} > 0)$ : US 0.01, EU 0.00, UK 0.02, DK 0.03, SCANDI 0.07, World 0.01.
- **Publication decay:** gross L/S -12.8% to -3.8% a year in 2013–19 against -15.9% to -1.6% in 2020–26.

## 10. What goes wrong, and how to fix it

**Method.** Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate  $p < 0.0020$ .

### 10.1 Diagnosis (design window 2013–2019)

| Universe | L/S gross / yr | Beta  | Beta drag / yr | CAPM alpha (gross) | Costs / yr | Long leg vs EW / yr | EW vs short leg / yr | Turnover / month | Worst months                            |
|----------|----------------|-------|----------------|--------------------|------------|---------------------|----------------------|------------------|-----------------------------------------|
| US       | -3.8%          | 0.27  | +3.5%          | -7.4%              | -1.7%      | -1.9%               | -2.0%                | 70%              | 2017-10 -8%, 2017-05 -8%, 2019-08 -7%   |
| EU       | -8.7%          | -0.06 | -0.6%          | -8.1%              | -1.7%      | -5.3%               | -3.5%                | 72%              | 2016-11 -9%, 2016-09 -8%, 2013-10 -7%   |
| UK       | -12.8%         | 0.18  | +2.1%          | -14.9%             | -1.7%      | -1.4%               | -11.4%               | 71%              | 2016-04 -15%, 2016-06 -10%, 2017-01 -9% |
| DK       | -10.9%         | -0.10 | -1.5%          | -9.4%              | -0.9%      | -4.2%               | -6.7%                | 37%              | 2013-05 -14%, 2018-06 -9%, 2015-12 -8%  |
| SCANDI   | -3.8%          | 0.03  | +0.4%          | -4.2%              | -1.3%      | -2.1%               | -1.7%                | 54%              | 2013-09 -11%, 2016-11 -11%, 2017-04 -8% |
| World    | -7.0%          | 0.21  | +2.3%          | -9.3%              | -1.8%      | -3.5%               | -3.5%                | 74%              | 2016-04 -8%, 2017-08 -7%, 2016-06 -7%   |

On average across the six universes the gross spread was -7.8% a year, of which the market exposure (beta +0.09) contributed +1.0%; the beta-adjusted spread (CAPM alpha) was -8.9%. Costs took 1.5% a year at 63% monthly turnover across both legs. The long leg beat the universe by -3.0% and the short leg lagged it by -4.8% a

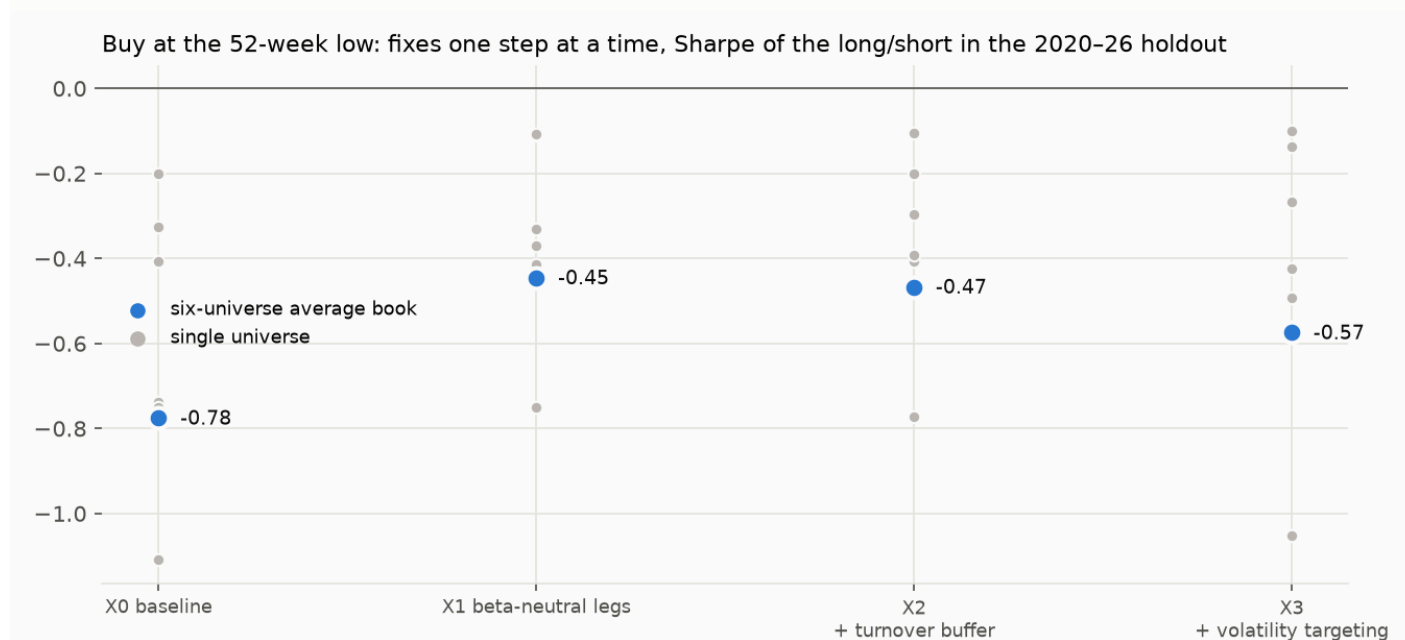
year, so most of the spread comes from the long side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +9.8%, +9.6%, +8.8% a year.

## 10.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

| Step                      | Design Sharpe (avg) | Δ design (p)    | Holdout Sharpe (book) | Δ holdout (p)   | Universes improved | Holdout return / yr | Holdout max DD | Holdout alpha vs EW (t) | Costs / yr |
|---------------------------|---------------------|-----------------|-----------------------|-----------------|--------------------|---------------------|----------------|-------------------------|------------|
| X0 baseline               | -0.72               | –               | -0.78                 | –               | –                  | -10.1%              | -57%           | -12.3% (-2.6)           | 1.6%       |
| X1 beta-neutral legs      | -0.58               | +0.18 (p 0.081) | -0.45                 | +0.33 (p 0.075) | 6 of 6             | -6.0%               | -52%           | -9.3% (-1.9)            | 1.7%       |
| X2 + turnover buffer      | -0.67               | -0.13 (p 0.908) | -0.47                 | -0.02 (p 0.547) | 3 of 6             | -5.1%               | -42%           | -8.0% (-2.0)            | 0.8%       |
| X3 + volatility targeting | -0.56               | +0.17 (p 0.354) | -0.57                 | -0.11 (p 0.942) | 3 of 6             | -6.0%               | -43%           | -8.8% (-2.3)            | 0.8%       |

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.

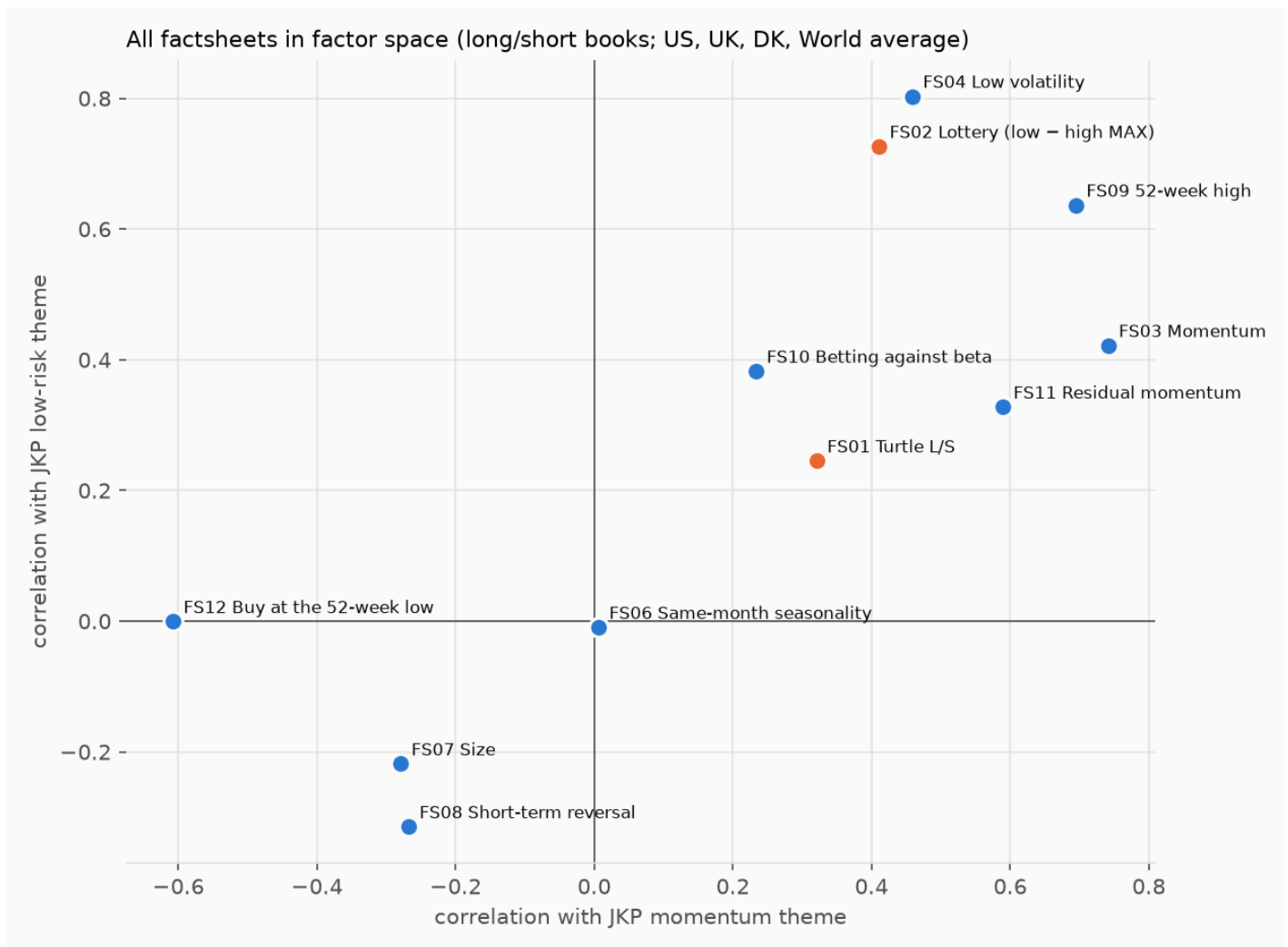


- **X1 beta-neutral legs:** +0.18 design, +0.33 holdout; helps in both windows but does not pass the gate.
- **X2 + turnover buffer:** -0.13 design, -0.02 holdout; hurts in both windows.
- **X3 + volatility targeting:** +0.17 design, -0.11 holdout; helps in one window and hurts in the other: not reliable.

**Where it ends:** holdout Sharpe -0.78 → -0.57, return -10.1% → -6.0% a year (six-universe book).

## 11. Classification: where does it fit?

| Dimension                         | Buy at the 52-week low                                                                                                                                        |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Series family                     | <b>Reversal &amp; bottom-fishing</b>                                                                                                                          |
| Academic style                    | Contrarian / bottom-fishing                                                                                                                                   |
| Signal                            | Price only (daily total returns)                                                                                                                              |
| Cross-section vs time series      | <b>Cross-sectional:</b> stocks ranked against each other each month                                                                                           |
| Horizon and turnover              | One month; 63% of the two legs replaced monthly                                                                                                               |
| Market exposure                   | L/S beta 0.09 to 0.30                                                                                                                                         |
| Payoff shape                      | Treynor–Mazuy $\gamma$ -1.14 to 1.93 (t -1.3 to 1.2); worst 10% of market months -2.2% to -0.6% a month, best 10% -0.8% to +1.6%                              |
| Nearest factor theme              | momentum (4 of 6 universes), WML (2 of 6 universes)                                                                                                           |
| Nearest library signals (returns) | Momentum 12-1 (6 universes), Last month return (4 universes), Residual momentum (3 universes)                                                                 |
| Economic rationale                | Folklore: mean reversion and value. Evidence: momentum and anchoring (George & Hwang 2004) predict continued weakness; losers carry high beta and volatility. |
| Publication                       | No academic origin; George & Hwang 2004 as the counter-claim; series A10 (Free Fallin')                                                                       |
| Where it fits                     | In the <b>momentum</b> cluster of the library; fully explained by its neighbours.                                                                             |

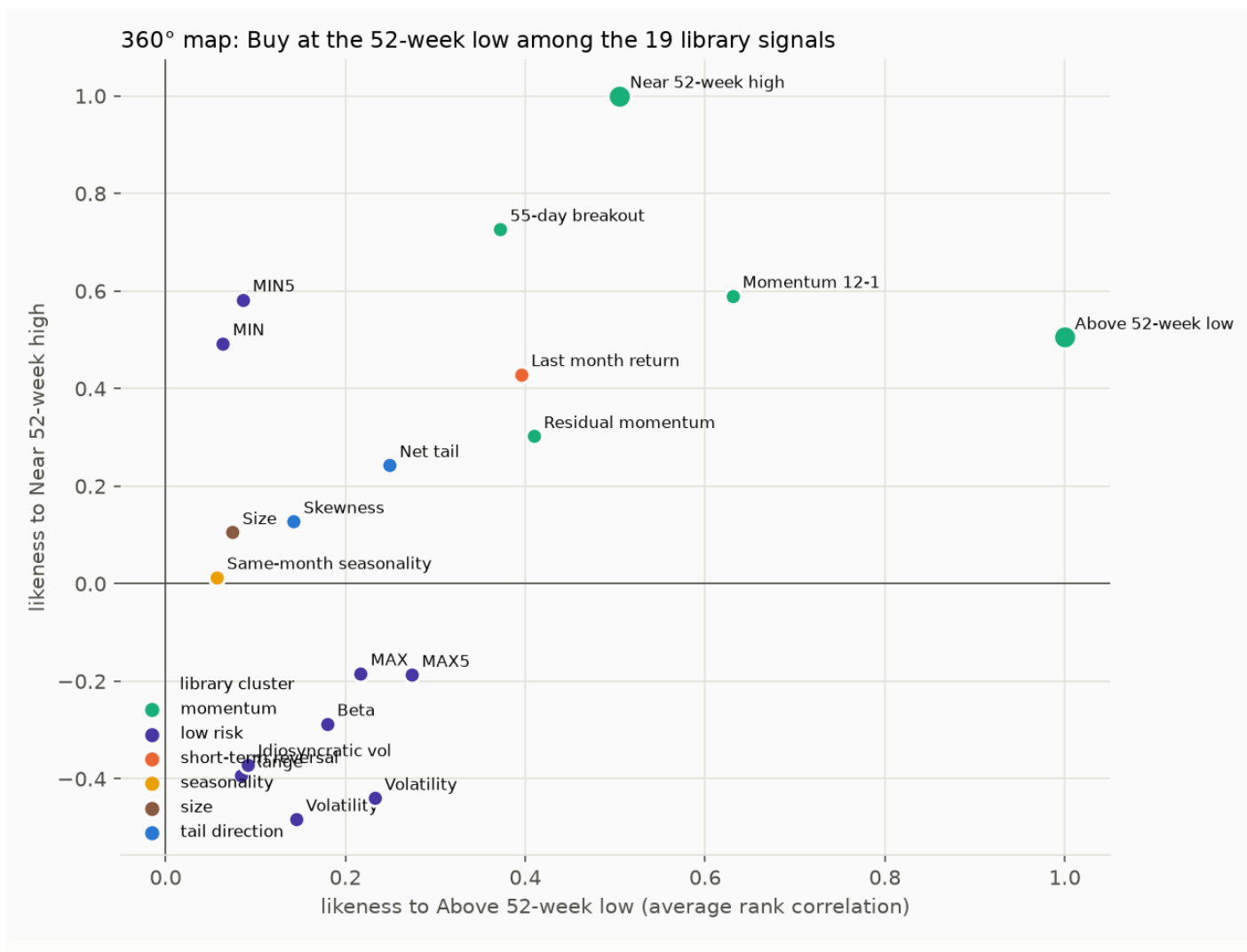


## 12. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Near 52-week high**.

| Signal (top minus bottom, raw sign)        | Cluster             | Likeness: stocks | Likeness: returns | Raw L/S, avg of 6 | CAPM alpha range | Universes alpha t ≥ +2 / ≤ -2 |
|--------------------------------------------|---------------------|------------------|-------------------|-------------------|------------------|-------------------------------|
| Above 52-week low ( <b>this strategy</b> ) | momentum            | +1.00            | +1.00             | +7.5%             | +5.2% to +13.6%  | 4 / 0                         |
| Momentum 12-1                              | momentum            | +0.63            | +0.60             | +7.2%             | +4.3% to +19.4%  | 5 / 0                         |
| Near 52-week high (mirror)                 | momentum            | +0.51            | +0.46             | +3.7%             | +9.3% to +22.3%  | 6 / 0                         |
| Last month return                          | short-term reversal | +0.40            | +0.46             | +1.8%             | -3.2% to +11.0%  | 2 / 0                         |
| Residual momentum                          | momentum            | +0.41            | +0.44             | +6.4%             | +8.0% to +12.2%  | 6 / 0                         |
| 55-day breakout                            | momentum            | +0.37            | +0.41             | +1.4%             | +4.1% to +13.0%  | 4 / 0                         |

| Signal (top minus bottom, raw sign) | Cluster        | Likeness: stocks | Likeness: returns | Raw L/S, avg of 6 | CAPM alpha range | Universes alpha $t \geq +2 / \leq -2$ |
|-------------------------------------|----------------|------------------|-------------------|-------------------|------------------|---------------------------------------|
| Net tail (MAX+MIN)                  | tail direction | +0.25            | +0.33             | -0.9%             | -5.2% to +5.2%   | 0 / 0                                 |
| MAX5                                | low risk       | +0.27            | +0.16             | +0.6%             | -11.3% to -3.5%  | 0 / 2                                 |
| Size                                | size           | +0.07            | +0.15             | +0.1%             | -3.8% to +9.9%   | 1 / 0                                 |
| MIN5                                | low risk       | +0.09            | +0.14             | -0.9%             | +2.3% to +13.7%  | 4 / 0                                 |
| MAX (best day)                      | low risk       | +0.22            | +0.12             | +0.1%             | -13.3% to -2.0%  | 0 / 2                                 |
| Skewness                            | tail direction | +0.14            | +0.12             | -0.4%             | -6.7% to +3.9%   | 0 / 2                                 |
| MIN (worst day)                     | low risk       | +0.06            | +0.12             | -0.7%             | +3.3% to +11.1%  | 4 / 0                                 |
| Idiosyncratic vol                   | low risk       | +0.09            | +0.07             | -3.6%             | -15.5% to -4.8%  | 0 / 4                                 |
| Volatility (252 days)               | low risk       | +0.23            | +0.04             | +2.4%             | -12.9% to -4.8%  | 0 / 3                                 |
| Volatility                          | low risk       | +0.14            | -0.02             | +1.1%             | -13.7% to -7.9%  | 0 / 5                                 |
| Range (MAX–MIN)                     | low risk       | +0.08            | -0.02             | -0.7%             | -15.4% to -4.8%  | 0 / 4                                 |
| Same-month seasonality              | seasonality    | +0.06            | +0.01             | -1.9%             | -9.4% to +6.6%   | 0 / 1                                 |
| Beta                                | low risk       | +0.18            | -0.01             | +4.4%             | -12.6% to -4.2%  | 0 / 4                                 |



**Spanning.** Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

| Universe | Alpha after neighbours / yr | t     | R <sup>2</sup> | Neighbours with  t  ≥ 2                                                                                                                           |
|----------|-----------------------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| US       | +0.0%                       | 0.02  | 0.70           | Momentum 12-1 +0.69 (+7.6), Residual momentum +0.28 (+2.5), Last month return +0.34 (+4.6), Net tail +0.36 (+2.9), Near 52-week high -0.45 (-5.0) |
| EU       | +2.8%                       | 1.07  | 0.59           | Momentum 12-1 +0.61 (+5.6), Last month return +0.46 (+6.2), Residual momentum +0.28 (+2.5), Near 52-week high -0.54 (-4.7)                        |
| UK       | -0.2%                       | -0.06 | 0.52           | Momentum 12-1 +0.43 (+3.8), Residual momentum +0.43 (+2.9), Near 52-week high -0.31 (-2.3), Last month return +0.43 (+4.4)                        |
| DK       | -1.3%                       | -0.49 | 0.55           | Momentum 12-1 +0.42 (+4.8), Last month return +0.34 (+3.7), Size +0.24 (+3.1)                                                                     |
| SCANDI   | +1.5%                       | 0.53  | 0.46           | Momentum 12-1 +0.30 (+2.4), 55-day breakout +0.22 (+2.0), Last month return +0.28 (+4.0), Residual momentum +0.30 (+2.5)                          |
| World    | -0.7%                       | -0.31 | 0.63           | Momentum 12-1 +0.66 (+4.9), Last month return +0.55 (+7.4), Near 52-week high -0.48 (-3.8)                                                        |

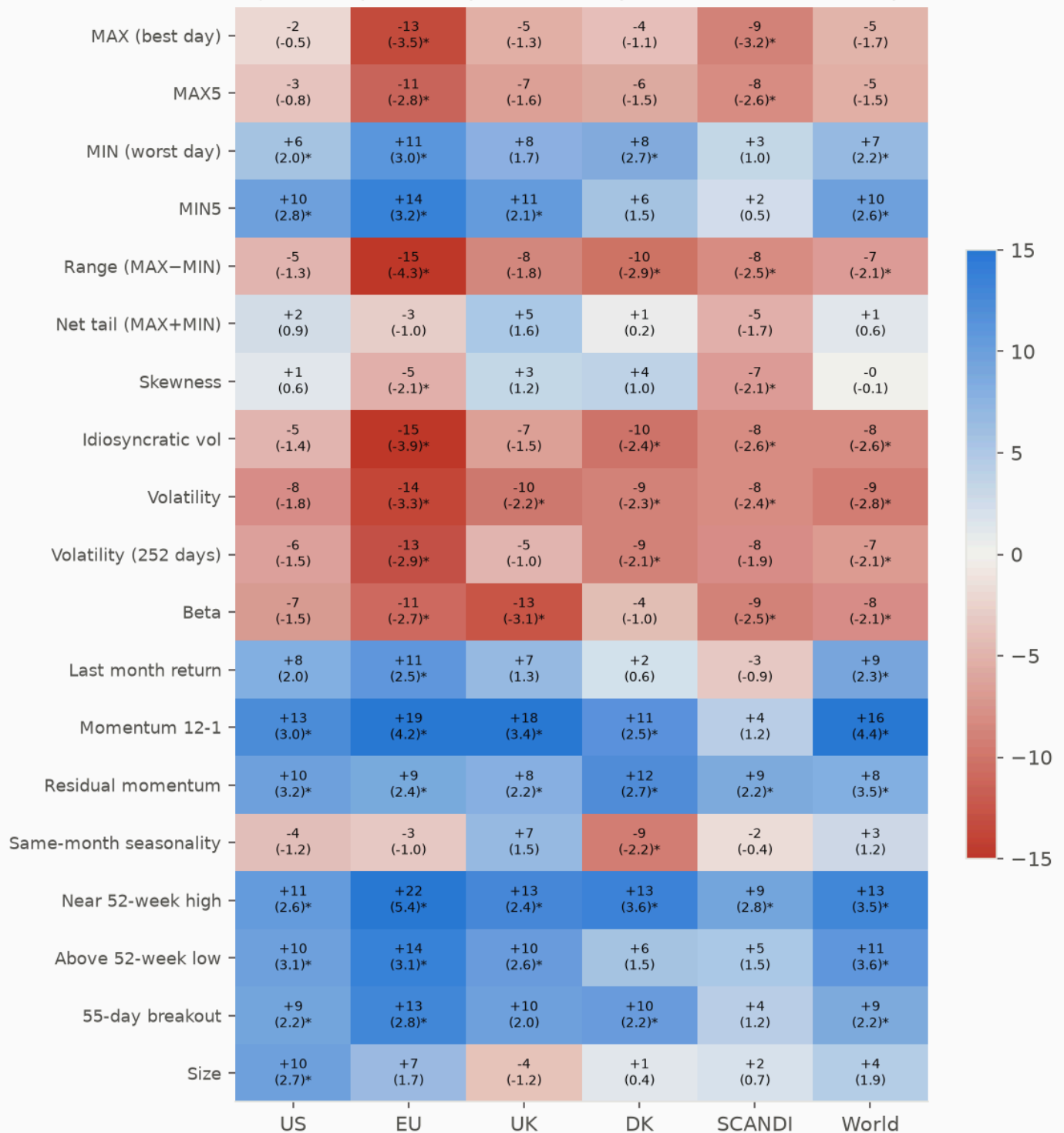
3×3 sort, next-month return a year  
(US, EU, UK, SCANDI, World averaged)

|                        |                       |        |                        |
|------------------------|-----------------------|--------|------------------------|
| low Above 52-week low  | 9.4%                  | 8.5%   | 9.8%                   |
| middle                 | 11.9%                 | 11.2%  | 11.8%                  |
| high Above 52-week low | 12.3%                 | 14.6%  | 13.5%                  |
|                        | low Near 52-week high | middle | high Near 52-week high |

Holding Near 52-week high fixed (down a column), moving from low to high Above 52-week low changes the return by +2.9, +6.1 and +3.7 points a year; holding Above 52-week low fixed (along a row), moving from low to high Near 52-week high changes it by +0.5, -0.1 and +1.2. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): +5% vs +12% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +7.5%, +9.8%, +9.6%, +8.8% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



## 13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.

3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.
4. **Not a registered trial.** Descriptive; the fix ladder is counted inside FSO3–FS12 (25 trials).

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## 15. Reproduce

code/data.py → signals.py, signals2.py, library.py → xs.py + analyse\_xs.py FS12 → build\_xs.py FS12.

Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.